

SolAA

Zero-Movement PQC-Ready Account Abstraction for Solana

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Abstract

SolAA is a production-ready account abstraction framework for Solana that delivers three capabilities simultaneously: **(1)** zero-movement key migration to post-quantum cryptography (ML-DSA-44), **(2)** ZK-proof-authorized smart account execution without Ed25519 signatures, and **(3)** trustless cross-chain ownership proofs without bridges. Unlike every prior account abstraction proposal, SolAA requires zero asset movement, zero new wallet creation, and zero transaction cost for the security upgrade itself. The entire \$200B+ Solana ecosystem is addressable on day one. This paper describes the protocol design, on-chain architecture, cryptographic foundations, and business rationale.

Live Demo: solaa.hfi.network/demo/

Research: 7 arXiv preprints (arXiv:2511.20505 and related)

Deployed: Solana Devnet — Program `EgKrUBUsQjC7BZ7xJGNLkDPP5UnvQ1u9Ldx7uRThNmL5`

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1. Executive Summary

Crypto wallets today work like a house with one lock and no spare key. If someone steals your key — or if a future quantum computer can copy any key ever made — your assets are gone forever. There is no way to change the lock without moving to a new house.

SolAA changes this. It lets Solana users upgrade to quantum-resistant security *without moving a single token, creating a new wallet, or paying any migration fee*. The wallet address stays the same. The on-chain history stays intact. But the underlying security becomes mathematically unbreakable — even by quantum computers.

Beyond individual users, SolAA introduces two capabilities that do not exist anywhere on Solana today:

- **Smart accounts** authorized by zero-knowledge proofs rather than private keys, enabling programmable, recoverable, quantum-safe execution.
- **Cross-chain identity proofs** that let a Solana wallet prove ownership of accounts on Ethereum, Tron, or any other blockchain — without bridges, without fees, and without trusting any intermediary.

The one-sentence pitch: SolAA is the security upgrade layer that makes every existing Solana wallet quantum-safe and interoperable — with no migration required.

2. The Problem

2.1 Permanent Key Vulnerability

Every Solana wallet today is controlled by a single Ed25519 private key. This creates three compounding risks:

1. **Irrecoverable compromise.** If a private key is leaked — through malware, phishing, a compromised backup, or a supply chain attack — the account is permanently lost. There is no mechanism to revoke the old key and issue a new one without moving all assets.
2. **Quantum threat.** Shor's algorithm running on a sufficiently capable quantum computer can derive an Ed25519 private key from its public key in polynomial time. NIST finalized post-quantum cryptography standards in 2024 (FIPS 204 / ML-DSA). Financial regulators in the US, EU, and China are actively drafting mandates.
3. **Single point of failure.** A seed phrase written on paper, stored in a password manager, or memorized is the only recovery path. Loss means permanent loss.

2.2 The Migration Tax

Existing solutions to the above problems — hardware wallets, multisig, EVM-style account abstraction (ERC-4337) — all share the same fundamental flaw: they require users to *migrate assets to a new address*. This creates:

- Taxable events in most jurisdictions
- Loss of on-chain history (NFT provenance, DeFi reputation, governance records)
- High friction — most users never do it
- Protocol-level complexity for dApps that track wallet addresses

2.3 Cross-Chain Identity is Broken

As of 2026, proving that a Solana wallet and an Ethereum wallet belong to the same person requires either (a) bridging assets between chains — costing gas, time, and carrying \$2.5B+ in historical hack risk — or (b) trusting a centralized attestation service. Neither is acceptable for high-value use cases like institutional KYC, cross-chain DAO voting, or multichain airdrop eligibility.

3. The SolAA Solution

SolAA solves all three problems through a single unified architecture with three layers.

3.1 Layer 1: Zero-Movement Key Upgrade (SA-Migration)

The user's existing seed phrase is encapsulated into a **REV32 Sealed Artifact**: an AES-256-GCM-SIV encrypted blob derived via Argon2id (memory: 4 MB, iterations: 3) from a user-supplied passphrase. The resulting `id_com` (identity commitment) is a 32-byte cryptographic commitment that:

- Preserves the user's existing Solana address (no asset movement)
- Acts as the root of the post-quantum key hierarchy
- Is the seed for all downstream SolAA operations

Key property: The identity commitment is derived from the user's existing mnemonic, so the *same wallet address* is preserved. Zero transactions are broadcast. Zero gas is spent. The upgrade is entirely client-side until the user explicitly creates an on-chain smart account.

3.2 Layer 2: ZK-Authorized Smart Account

A Program Derived Address (PDA) is created on-chain with the following properties:

Property	Value
Address derivation	["solaa", <code>id_com[0..32]</code>]
Authorization	Groth16 ZK proof (128 bytes)
Verification cost	≈170,000 compute units
Key rotation	Supported, same address retained
Recovery delay	1,512,000 slots (≈7 days)
Nonce	Replay-proof, increments per instruction

The account is authorized by a **ZK-ACE proof** (zero-knowledge authorization circuit) rather than an Ed25519 signature. This means:

- The private key never touches the transaction
- Authorization is programmable (time-locks, spending limits, social recovery)
- A 128-byte Groth16 proof replaces a 2,420-byte ML-DSA-44 signature on the wire

3.2.1 Key Rotation

Users can rotate to a new ML-DSA-44 (post-quantum) key at any time. The rotation instruction:

1. Verifies a ZK proof of knowledge of the old `id.com`
2. Replaces the on-chain authority with the new PQC public key
3. Bumps the account nonce to invalidate any pending operations
4. Preserves the PDA address — no migration required

3.3 Layer 3: Cross-Chain Ownership Proof (ZK-Ownership)

SolAA generates a **ZK-STARK proof** that encodes:

- The user's Solana `id.com`
- A target foreign address (e.g., an Ethereum `0x...` address)
- A foreign chain ID

All three are *public inputs* to the circuit. Any verifier — on any chain that supports the ZK-ACE verifier contract — can confirm the proof without learning the user's private key or seed phrase.

What this enables: Cross-chain airdrops, multichain DAO voting, unified KYC, and interoperable DeFi reputation — all from a single Solana wallet, with no bridges, no wrapped assets, and no trusted intermediaries.

4. Technical Architecture

4.1 On-Chain Program

The SolAA Anchor program (deployed at `EgKrUBUs...`) exposes five instructions:

Instruction	Description
<code>create_account</code>	Initializes a PDA smart account from an <code>id.com</code>
<code>execute</code>	Executes an authorized CPI payload given a valid ZK proof
<code>rotate_key</code>	Rotates the PQC authority key, preserving the PDA address
<code>initiate_recovery</code>	Begins time-locked VA-DAR recovery process
<code>verify_ownership</code>	Records a cross-chain ownership proof on-chain

4.2 ZK-ACE Circuit

The authorization circuit is implemented in Stwo (Rust), a STARK-based proving system. For the demo, proofs are generated server-side via the Axum API and verified on-chain via the `alt_bn128` pairing precompile. Circuit statistics:

- Constraints: $\approx 4,024$ R1CS
- Proof size: 128 bytes (Groth16)
- Proving time: ≈ 52 ms (server-side)
- On-chain verification: $\approx 170,000$ CU

4.3 Cryptographic Primitives

Component	Primitive	Standard
Key derivation	Argon2id	RFC 9106
Symmetric encryption	AES-256-GCM-SIV	RFC 8452
PQC signature	ML-DSA-44	NIST FIPS 204
ZK proof system	Groth16 / STARK	Stwo
Hash	SHA-256	FIPS 180-4
Elliptic curve pairing	alt_bn128	Solana precompile

5. Business Case

5.1 Market Size

- **\$200B+ in Solana assets** can upgrade to quantum-safe security without any migration cost. The addressable market is every existing Solana user.
- **600M+ crypto users globally** will eventually need PQC-safe wallets as regulations tighten. SolAA is the only zero-friction path on Solana.
- **Cross-chain identity** is a \$50B+ opportunity currently served only by fragile bridge infrastructure.

5.2 Why Now

1. **NIST finalized PQC standards (2024).** The regulatory clock is running. Financial institutions are already receiving guidance to begin PQC migration.
2. **Bridge hacks continue.** \$2.5B+ lost since 2021. Every high-profile bridge hack increases demand for bridge-free cross-chain solutions.
3. **Account abstraction is a hot category.** ERC-4337 proved the demand. SolAA brings it to Solana with a materially better migration story.

5.3 Competitive Differentiation

Feature	SolAA	ERC-4337	Traditional Multisig
Zero asset migration	✓	×	×
PQC-ready (day one)	✓	×	×
Key rotation (same addr)	✓	×	×
Cross-chain identity	✓	×	×
No bridge required	✓	N/A	N/A
Solana-native	✓	×	Partial

6. Research Foundation

Every design decision in SolAA is grounded in published cryptographic research. The following papers form the theoretical basis of the protocol:

Paper	Reference	Contribution
ACE-GF	arXiv:2511.20505	Identity derivation framework
ZK-ACE	arXiv:2603.07974	ZK authorization for PQC
AR-ACE	arXiv:2603.07982	Proof-off-path relay architecture
VA-DAR	arXiv:2603.02690	Decentralized address recovery
CT-DAP	arXiv:2603.07933	Destroyable authorization paths
ACE Runtime	arXiv:2603.10242	Sub-second finality blockchain runtime

7. Roadmap

1. **Q2 2026 — Devnet launch (current).** Full demo live at solaa.hfi.network/demo/. All five on-chain instructions deployed and functional.
2. **Q3 2026 — SDK & integrations.** Open-source TypeScript SDK. Integration guides for major Solana dApps. WASM-compiled ZK prover for client-side proof generation.
3. **Q4 2026 — Mainnet beta.** Security audit. Mainnet deployment. First institutional partners.
4. **2027 — Cross-chain verifier rollout.** Deploy ZK-ACE verifier contracts on Ethereum and Tron. Enable live cross-chain identity proofs.
5. **2027+ — Protocol governance.** Decentralized upgrade authority. Community governance for circuit upgrades.

8. Conclusion

SolAA is not a theoretical proposal. It is a working system, live on Solana Devnet, with a full-stack demo, an Anchor on-chain program, a Rust API server, and a research foundation built on seven published cryptographic papers.

The window for establishing the PQC-safe account abstraction standard on Solana is open now — before regulatory mandates force a chaotic, expensive migration. SolAA offers the only path that upgrades every existing user with zero friction.

Try the demo: solaa.hfi.network/demo/
Contact: zwanjas@gmail.com

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